

Rapid-Acting Insulin CDS — Comprehensive Parameter Grid

Every configured constant and limit, extracted from the deployed calculator (CDS_CONFIG). U-10 Dilute parameters paired with their U-100 Standard counterparts.

Constant	Value	Unit	Action Type	Notes
INSULIN : CARBOHYDRATE RATIO (order entry field)				
U-100 I:C minimum	2	g / Unit	Order hard limit	Below min snaps up to 2; warning fires; cannot sign out of range.
U-100 I:C maximum	100	g / Unit	Order hard limit	Above max snaps down to 100; warning fires.
U-10 Dilute I:C minimum	1.5	g / Syringe Unit	Order hard limit	Below min snaps up to 1.5; warning fires; cannot sign out of range.
U-10 Dilute I:C maximum	40	g / Syringe Unit	Order hard limit	Above max snaps down to 40; warning fires.
CORRECTION FACTOR / INSULIN SENSITIVITY FACTOR (order entry field)				
U-100 CF minimum	8	mg/dL / Unit	Order hard limit	Below min snaps up to 8; warning fires.
U-100 CF maximum	300	mg/dL / Unit	Order hard limit	Above max snaps down to 300; warning fires.
U-100 CF advisory threshold	> 150	mg/dL / Unit	Order warn threshold	Soft advisory: verify sensitivity factor; prescriber may acknowledge and proceed.
U-10 Dilute CF minimum	10	mg/dL / Syringe Unit	Order hard limit	—
U-10 Dilute CF maximum	200	mg/dL / Syringe Unit	Order hard limit	—
TARGET BLOOD GLUCOSE (order entry field, age-stratified)				
U-100 Target BG min — < 5 years	130	mg/dL	Order hard limit	Age band selected at order entry.
U-100 Target BG max — < 5 years	200	mg/dL	Order hard limit	—
U-100 Target BG min — 5–11 years	70	mg/dL	Order hard limit	—
U-100 Target BG max — 5–11 years	180	mg/dL	Order hard limit	—
U-100 Target BG min — ≥ 12 years	70	mg/dL	Order hard limit	—
U-100 Target BG max — ≥ 12 years	150	mg/dL	Order hard limit	—
U-10 Dilute Target BG min — ≤ 12 months	120	mg/dL	Order hard limit	Dedicated U-10 band; wider range for infai physiology.
U-10 Dilute Target BG max — ≤ 12 months	300	mg/dL	Order hard limit	—
DOSE ROUNDING (order setting)				
U-100 Rounding increment (toggle)	0.5 or 1	Unit	Order setting	Prescriber toggle; default 0.5. Applied to summed total.
U-10 Dilute Rounding increment (fixed)	1	Syringe Unit	Order setting	Fixed — no toggle. Always rounds up to nearest integer Syringe Unit.
CARBOHYDRATES EATEN (calculator input)				
U-100 Carbs entry minimum	0	g	Calculator hard limit	Entry rejected below 0.
U-100 Carbs entry maximum	300	g	Calculator hard limit	Entry rejected above 300.
U-100 Carbs formula cap	250	g	Calculator warn threshold	Entry 251–300 informs RN and substitutes 250 g in prandial formula; entered value retained.
U-10 Dilute Carbs entry minimum	0	g	Calculator hard limit	—

U-10 Dilute Carbs entry maximum	99	g	Calculator hard limit	Entry ceiling.
CURRENT BLOOD GLUCOSE (calculator input — applies to both concentrations)				
Blood glucose entry minimum	1	mg/dL	Calculator hard limit	Entry rejected below 1.
Blood glucose entry maximum	1000	mg/dL	Calculator hard limit	Entry rejected above 1000.
Blood glucose formula cap	600	mg/dL	Calculator warn threshold	BG > 600 informs RN and substitutes 600 in correction formula; prandial unaffected.
Blood glucose hypo threshold	< 70	mg/dL	Calculator Stop Threshold	All dose components suppressed; rescue guidance; new entry required for any subsequent dose.
CALCULATOR POLICY (behavior constants — both concentrations)				
Insulin-on-board lookback window	3	hours	Calculator Policy	Rapid-acting insulin within window holds the correction component (anti-stacking); prandial preserved; nurse can override.
Rounding direction	ceiling	—	Calculator Policy	Always rounds up after summation.

Action type definitions

- Order hard limit** Out-of-range value at order entry snaps to the boundary with a warning; order cannot be signed out of range.
- Order warn threshold** Soft advisory at order entry; prescriber may acknowledge and proceed.
- Calculator hard limit** Entry rejected by the calculator input field; value cannot be entered.
- Calculator warn threshold** Calculator informs the nurse and substitutes a capped value in the formula; entered value retained for verification.
- Calculator Stop Threshold** Calculator suppresses all dose output and presents rescue/clinical guidance.
- Order setting** Prescriber-configured behavior carried on the order (not a limit or threshold).
- Calculator Policy** Fixed calculator behavior constant (not a limit, threshold, or order field).